

```
> stemCompletion(c("op", "so"), feed)
      op      so
```

```
"open-source" "software"
```

removeNumbers

You can remove numbers using the *removeNumbers()* function. For example:

```
> lines[3]
[1] "Products include permission to use the source code,[1]
design documents,[2] or content of the product. The"
> removeNumbers(lines[3])
```

```
[1] "Products include permission to use the source code,[
design documents,[ or content of the product. The"
```

removePunctuation

The punctuation marks in the text can be removed using the *removePunctuation()* function, as demonstrated below:

```
> lines[12]
[1] "rise of the Internet.[10] The open-source software
movement arose to clarify copyright, licensing, domain,"
> removePunctuation(lines[12])
```

```
[1] "rise of the Internet10 The opensource software movement
arose to clarify copyright licensing domain"
```

The following arguments are accepted by the *removePunctuation()* function:

Argument	Description
x	A character vector or input text
preserve_intra_word_con-	Logical value to keep intra-
tractions	word contractions
preserve_intra_word_	Logical value to keep intra-
dashes	word dashes
ucp	Logical value to use Uni-
	code character properties
...	Additional arguments to be
	passed

removeWords

The *removeWords()* function removes words from the text, as illustrated below:

```
> lines[24]
[1] "instance, in the early years of automobile development a
group of capital monopolists owned the rights to a"
```

```
> removeWords(lines[24], stopwords("en"))
[1] "instance,  early years  automobile development  group

capital monopolists owned rights "
```

findFreqTerms

A term-document matrix can be constructed using the *TermDocumentMatrix()* function. You can then use the *findFreqTerms()* function to list the words between a lower and upper frequency, as follows:

```
> tdm <- TermDocumentMatrix(lines)
> findFreqTerms(tdm, 3, 5)
[1] "modification" "include" "decentralized"
"encourages"
[5] "main" "principle" "production,"
"blueprints,"
[9] "public." "began" "code."
"limitations"
[13] "promotes" "became" "redistribution"
"widely"
[17] "developers" "gained" "hold" "producers"
[21] "domain," "general" "program"
"design."
...
```

The *findFreqTerms()* function accepts the following arguments:

Argument	Description
x	A document-term matrix
lowfreq	Lower bound numeric frequency
highfreq	Upper bound numeric frequency

findMostFreqTerms

A term frequency vector can be generated using the *termFreq()* function, and you can use it to find the most frequently used words in the document with the *findMostFreqTerms()* function, as shown below:

```
> tf <- termFreq(lines)
> findMostFreqTerms(tf)
the      and      open      source open-source      for

520      296      222      174      161      123
```